

consteval blocks

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§ 1 Introduction

Several proposals that produce side effects as part of constant evaluation are in flight. That includes [\[P2996R2\]](#) (“Reflection for C++26”) and [\[P2758R2\]](#) (“Emitting messages at compile time”). Such a capability, in turn, quickly gives rise to the desire to evaluate such constant expressions in declarative contexts.

Currently, this effect can be shoe-horned into `static_assert` declarations, but the result looks arcane. For example, P2996 contains the following code in an example:

```
#include <meta>

template<typename... Ts> struct Tuple {
    struct storage;
    static_assert(
        is_type(define_class(^storage, {data_member_spec(^Ts)...})));
    storage data;

    Tuple(): data{} {}
    Tuple(Ts const& ...vs): data{ vs... } {}
};
```

Here, `define_class(...)` is a constant expression with a side-effect that we want to evaluate before parsing the member declaration that follows. It works, but it is somewhat misleading: We're not really trying to assert anything; we just want to force that evaluation.

We therefore propose a simple, intuitive syntax to express that we simply want to constant-evaluate a bit of code wherever a `static_assert` declaration could appear by enclosing that code in `consteval { ... }` (a construct we'll call a `consteval` block).

§ 2 Proposal

Formally, we propose that a construct of the form

```
consteval {  
    statement-seqopt  
}
```

is equivalent to:

```
static_assert(  
    (  
        []() -> void consteval {  
            statement-seqopt  
        }(),  
        true  
    )  
);
```

The `static_assert` condition is a *comma-expression* with the first operand an immediately-invoked `consteval` lambda.

Note that this allows a plain `return;` statement or even a `return f();` statement where `f()` has type ``void`. We could go out of our way to disallow that, but we cannot find any benefit in doing so.

With the feature as proposed, the example above becomes:

Status Quo	Proposed
<pre>#include <meta> template<typename... Ts> struct Tuple { struct storage; static_assert(is_type(define_class(^storage, {data_member_spec(^Ts)...}))); storage data; };</pre>	<pre>#include <meta> template<typename... Ts> struct Tuple { struct storage; consteval { define_class(^storage, {data_member_spec(^Ts)...}); } storage data; };</pre>

In this example, there is just a single expression statement being evaluated. However, we are anticipating reflection code where more complex statement sequences will be used (you can see some examples in previous papers, e.g. [P1717R0] [P2237R0]).

We did consider other syntax variations such as

- `@eval expression;`
- `@ expression;`
- `@ statement;`

but found those alternatives less general and not worth the slight improvement in brevity.

§ 3 Implementation Status

The Lock3 implementation of reflection facilities based on [P1240R2] (and other papers) includes this feature. The EDG front end is expected to add this feature shortly as part of its reflection extensions.

§ 4 Wording

[Editor's note: The simplest way to do the wording is to add a `consteval` block as a *kind of static_assert-declaration*. That's the minimal diff. However, it's kind of weird to say that a `consteval` block literally is a `static_assert` - even if we specify the former in terms of the latter. So we'd rather take a few more words to get somewhere that feels more sensible. Plus this change reduces a lot of duplication between *empty-declaration* and *static_assert-declaration*, which are treated the same in a lot of places anyway.]

Change 6.2 [basic.def]/2:

- 2 Each entity declared by a *declaration* is also defined by that *declaration* unless:
- (2.1) — it declares a function without specifying the function's body ([dcl.fct.def]),

- (2.2) — [...]
- (2.13) — it is a ~~static_assert-declaration~~ **vacant-declaration** ([dcl.pre]),
- (2.14) — it is an ~~attribute-declaration~~ ([dcl.pre]),
- (2.15) — ~~it is an empty-declaration~~ ([dcl.pre]),
- (2.16) — [...]

Change 9.1 [dcl.pre]:

```

name-declaration:
    block-declaration
    nodeclspec-function-declaration
    function-definition
    friend-type-declaration
    template-declaration
    deduction-guide
    linkage-specification
    namespace-definition
-   empty-declaration
    attribute-declaration
    module-import-declaration

block-declaration:
    simple-declaration
    asm-declaration
    namespace-alias-definition
    using-declaration
    using-enum-declaration
    using-directive
-   static_assert-declaration
    alias-declaration
    opaque-enum-declaration
+   vacant-declaration

+   vacant-declaration:
+       static_assert-declaration
+       empty-declaration
+       consteval-block-declaration

static_assert-declaration:
    static_assert ( constant-expression ) ;
    static_assert ( constant-expression , static_assert-message ) ;

+   consteval-block-declaration:
+       consteval compound-statement

```

And then after 9.1 [dcl.pre]/13:

13 *Recommended practice:* When a *static_assert-declaration* fails, [...]

* The *constexpr-block-declaration*

```
constexpr compound-statement
```

is equivalent to

```
static_assert(([]() -> void constexpr compound-statement(), true));
```

[Note 1: Such a *static_assert-declaration* never fails. — end note]

14 An *empty-declaration* has no effect.

Adjust the grammar in 11.4.1 [class.mem.general] and the rule in p3:

```
member-declaration:
    attribute-specifier-seqopt decl-specifier-seqopt member-declarator-
        listopt;
    function-definition
    friend-type-declaration
    using-declaration
    using-enum-declaration
-   static_assert-declaration
+   vacant-declaration
    template-declaration
    explicit-specialization
    deduction-guide
    alias-declaration
    opaque-enum-declaration
-   empty-declaration
```

3 A *member-declaration* does not declare new members of the class if it is

- (3.1) — a friend declaration ([class.friend]),
- (3.2) — a *deduction-guide* ([temp.deduct.guide]),
- (3.3) — a *template-declaration* whose declaration is one of the above,
- (3.4) — a ~~*static_assert-declaration*~~,
- (3.5) — a *using-declaration* ([namespace.udecl]) , or
- (3.6) — ~~an *empty-declaration*~~: a *vacant-declaration*.

And similar in 11.5.2 [class.union.anon]/1. [Editor's note: This refactor allows putting in an *empty-declaration* into an anonymous union, which is kind of a consistency drive by with other classes.]

1 [...] Each *member-declaration* in the *member-specification* of an anonymous union shall either define one or more public non-static data members or be a ~~*static_assert-declaration*~~ *vacant-declaration*. [...]

§ 4.1 Feature-Test Macro

Add to the table in 15.11 [\[cpp.predefined\]](#):

```
__cpp_consteval      202211L  
+ __cpp_consteval_block 2024XXL  
__cpp_constinit      201907L
```

§ 5 References

[P1240R2] Daveed Vandevoorde, Wyatt Childers, Andrew Sutton, Faisal Vali. 2022-01-14. Scalable Reflection.

<https://wg21.link/p1240r2>

[P1717R0] Andrew Sutton, Wyatt Childers. 2019-06-17. Compile-time Metaprogramming in C++.

<https://wg21.link/p1717r0>

[P2237R0] Andrew Sutton. 2020-10-15. Metaprogramming.

<https://wg21.link/p2237r0>

[P2758R2] Barry Revzin. 2024-02-15. Emitting messages at compile time.

<https://wg21.link/p2758r2>

[P2996R2] Barry Revzin, Wyatt Childers, Peter Dimov, Andrew Sutton, Faisal Vali, Daveed Vandevoorde, Dan Katz. 2024-02-15. Reflection for C++26.

<https://wg21.link/p2996r2>